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E1. Looking at Towers (easy version)

time limit per test: 4 seconds

memory limit per test: 512 megabytes

This is the easy version of the problem. The only differences between the easy and the difficult version are the constraints on t and n .

Consider a row of m towers; the height of the i -th tower in the row is h_i .

If you look at this row of towers from the left, you see all towers that are strictly higher than all towers before them. Similarly, if you look at this row of towers from the right, you see all towers that are strictly higher than all towers after them. For example, if the towers have heights $[3, 5, 5, 7, 4, 6, 7, 2, 4]$, then:

- when looking from the left, you see towers with heights 3, 5 and 7;
- when looking from the right, you see towers with heights 7 and 4.

Let $L(h)$ be the set of heights you see from the left, and $R(h)$ be the set of heights you see from the right when the sequence of heights is h . In the example above, $L(h) = \{3, 5, 7\}$, and $R(h) = \{4, 7\}$.

You are given a sequence $a_1, a_2, \dots, a_n$. Your task is to calculate the number of subsequences of a such that $L(a) = L(a')$ and $R(a) = R(a')$, where a' is the subsequence you consider. Two subsequences are different if indices of chosen elements are different.

Input

The first line contains one integer t ($1 \leq t \leq 100$) — the number of test cases.

**Educational Codeforces Round
182 (Rated for Div. 2)**

Contest is running

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Contestant



→ **Submit?**

Language: Python 3.13.2 
Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Submit

Each test case consists of two lines:

- the first line contains one integer n ($1 \leq n \leq 5000$);
- the second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

Additional constraint on the input: the sum of n over all test cases does not exceed 5000.

Output

For each test case, print one integer — the number of subsequences a' of the given sequence a such that $L(a) = L(a')$ and $R(a) = R(a')$. Since it might be huge, print it modulo 998244353.

Example

input	Copy
5 5 4 2 4 8 3 5 1 2 3 2 1 6 1 2 3 3 2 1 9 3 5 5 7 4 6 7 2 4 1 10	
output	Copy
5 1 3 51 1	

Note

In the first example, $L(a) = \{4, 8\}$, $R(a) = \{3, 8\}$. The subsequences included in the answer are:

- $[4, 8, 3]$ (the 1-st, the 4-th and the 5-th element);
- $[4, 8, 3]$ (the 3-rd, the 4-th and the 5-th element);
- $[4, 2, 8, 3]$ (the 1-st, the 2-nd, the 4-th and the 5-th element);
- $[4, 4, 8, 3]$ (the 1-st, the 3-rd, the 4-th and the 5-th element);
- $[4, 2, 4, 8, 3]$ (the whole sequence).

In the second example, the only valid subsequence is the given sequence itself.

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